

Mayank Balmoori

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Summary

Full-stack software engineer with hands-on experience building and shipping production systems across healthcare data infrastructure, real-time collaborative applications, and blockchain/Web3 platforms. Proficient in React, Node.js, TypeScript, WebSockets, and REST API design. Demonstrated ability to deliver end-to-end features independently in fast-paced startup and enterprise environments. Strong foundation in data structures, algorithms, system design, and computer science fundamentals.

Education

Vellore Institute of Technology (VIT)

Bachelor of Technology in Computer Science and Engineering

Vellore, India

Aug. 2023 – May 2027 (Expected)

- CGPA: 8.97 / 10 Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming

Experience

Software Engineer Intern

Providence India

May 2026 – Present

Hyderabad, India

- Redesigned a core file-transfer capability of the Data Exchange Platform (DXP) — a healthcare job-management and secured-transfer application — replacing full-file buffering with chunk-based streaming to hold memory constant regardless of file size and remove a ~2 GB ceiling that had blocked 50 GB+ transfers under production load.
- Implemented resumable transfers with checkpoint-based recovery for FTPS and SFTP, using REST offsets to resume on non-seekable FTPS streams so interrupted jobs restart from the last committed chunk instead of re-transferring the full file; added atomic .tmp-staging with rename to guarantee no partial or corrupt files reach the destination.
- Built a T-SQL cron engine (5 stored procedures, 3 scalar functions) that scans 2,500+ scheduled jobs against a deployment window in seconds, supporting special cron syntax (L, W, #) with per-job error isolation; shipped to live production within one week and adopted as the standard pre-release job-impact tool.
- Added a configurable per-connection host-key enforcement toggle across DTOs, mappers, and transfer handlers, letting operators safely bypass SFTP host-key checks during server migrations while preserving captured host-key data and auto-clearing stale conflict flags — eliminating unnecessarily blocked jobs.
- Documented designs via Architecture Decision Records (ADRs) and READMEs; worked in an Agile team with sprints, stand-ups, and full code reviews.

Full Stack Intern

OriginHash THub

May 2025 – June 2025

Hyderabad

- Architected and delivered 70% of the Web2 platform within the first month — full-stack development across a React frontend and Node.js/Express backend with MongoDB.
- Designed scalable REST APIs for core data operations; collaborated directly with the founder to drive product requirements and key architectural decisions; shipped end-to-end features from design to deployment independently.

Projects

Canvix | Live Demo | GitHub

React, TypeScript, Canvas API, WebSockets, Node.js, Turborepo

Feb. 2026 – Present

- Built a real-time collaborative whiteboard supporting multi-user drawing, shape manipulation, room-based collaboration, and live synchronization using WebSockets.
- Designed a dual-canvas rendering architecture with separate preview and committed layers, enabling smooth drag, resize, and rotation operations without expensive full-scene re-renders.
- Implemented an event-based undo/redo system tracking shape operations instead of storing canvas snapshots, reducing memory overhead while synchronizing changes across all connected clients in real time.
- Structured as a Turborepo monorepo with shared TypeScript packages and centralized type definitions across the React frontend and Node.js backend; open-sourced on GitHub.

HD Wallet Web App (ETH + SOL) | GitHub

Next.js, TypeScript, Ethers.js, Solana Web3.js, BIP39, Blockchain

Sept. 2025 – Oct. 2025

- Built a multi-chain Hierarchical Deterministic (HD) wallet supporting Ethereum and Solana account generation from a single BIP39 mnemonic phrase.
- Implemented the complete wallet generation pipeline: mnemonic generation, PBKDF2 seed derivation, BIP32 master key creation, and BIP44 chain-specific account derivation.
- Integrated Ethers.js and Solana Web3.js to generate, manage, and display wallet addresses and private keys; documented the build in a technical write-up on Hashnode.

ChatApp | GitHub

MERN Stack, WebSockets, Docker, MongoDB

Mar. 2025 – May 2025

- Developed a real-time chat platform supporting anonymous room-based communication using WebSockets and event-driven message delivery.
- Implemented automatic room cleanup and ephemeral messaging — room state and messages are removed when the final participant disconnects.
- Containerized the full application with Docker for consistent local development and production-ready deployment; backend built with Node.js, Express.js, and MongoDB.

Technical Skills

Languages: JavaScript, TypeScript, Java, C, C++, SQL

Familiar / Working Experience: C#, ASP.NET, T-SQL, FluentFTP, SSH.NET

Frontend: React.js, Next.js, Tailwind CSS, Canvas API

Backend: Node.js, Express.js, REST APIs, WebSockets

Databases: PostgreSQL, MongoDB

Cloud & DevOps: Docker, AWS, Git, GitHub, Turborepo, Monorepos

Blockchain: Ethereum, Solana, Ethers.js, Solana Web3.js, BIP39, BIP44

Software Engineering: Agile, OOP, Distributed Systems, API Development, System Design

Computer Science: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS